

SoC'26- Dynamic Pricing Using RL

Week 1

Game:

A game involves 3 pieces: Players- the decision makers, Strategies- the choices each player can make and Payoffs- what each player gets depending on everyone's choices in the game and not just theirs.

Prisoner's Dilemma:

Two suspects are interrogated separately. Each player can Confess or Stay Silent. Payoffs are as follows:

Player A \ Player B	Confess	Stay Silent
Confess	5,5	0,10
Stay Silent	10,0	1,1

If the other person stays silent, what's the best move? Confess ($0 < 1$). If the other person confesses, what's your best move? Confess ($5 < 10$). Confess is better no matter what the other person does. That is a dominant strategy. Both players reason this way, both confess, both get 5 years. But if they'd both stayed silent, they'd each get 1 year. Individually rational choices leads to collectively bad outcome. Here Stay Silent for both is dominated strategy.

Nash Equilibrium:

A set of strategies (one per player) is a Nash equilibrium(NE) if no player can do better by switching, *given what everyone else is doing*.

Formally: strategy profile $(s_1^*, s_2^*, \dots, s_n^*)$ is a NE if for every player i :

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \text{ for all alternative } s_i$$

NE is **not** about everyone getting their best possible outcome. It is about no one having a unilateral incentive to deviate. The confess/confess outcome above is a Nash equilibrium even though it's bad for both players, neither can improve by switching alone.

Bertrand Model:

Assume two firms sell an identical product. Customers buy from whoever's cheaper (split evenly if tied). Each firm picks a price p_1, p_2 simultaneously. Marginal cost of production is c for both.

Logic, step by step:

- If $p_1 > p_2$: all customers go to firm 2. Firm 1 gets zero profit.
- If $p_1 < p_2$: all customers go to firm 1.

- So if you're firm 1 and firm 2 is charging some price above cost, you want to charge slightly less than them and capture the whole market.
- But firm 2 thinks the same thing about you.
- This undercutting continues... until price equals marginal cost, $p_1 = p_2 = c$.

At $p = c$, neither firm can profitably undercut further (they'd sell below cost) and neither wants to raise price (they'd lose all customers). That's the Nash equilibrium: price = marginal cost, profit = zero, even with just two competitors. This is the famous "Bertrand paradox", competition this sharp drives profit to nothing almost immediately.

Stackelberg

Bertrand and Cournot assume simultaneous moves. Stackelberg breaks that: one firm (the leader) sets its price/quantity first, the other (the follower) observes and reacts. The leader has an advantage. It can commit to an aggressive position knowing the follower will optimize given that commitment, via backward induction

The Evolution of Trust

A one-shot Prisoner's Dilemma ends in mutual defection (like Bertrand collapsing to cost). A repeated game opens the door to cooperation — players can reward/punish based on past behaviour. **Tit-for-Tat** is forgiving but not exploitable. This is the mechanism behind algorithmic collusion: agents that interact repeatedly can learn punishment strategies that sustain prices above the Bertrand price, without communicating or being told to collude

The Calvano paper

The researchers built Q-learning agents (a basic reinforcement learning algorithm, no communication, no explicit instructions to collude) and dropped them into a repeated Bertrand-style pricing game. Classical theory says they should converge to $p = c$, zero profit, exactly like we derived above. Instead, the algorithms consistently learn to charge supracompetitive prices, prices above the Bertrand level, without communicating with one another. The agents learn punishment strategies, if one agent undercuts, the others retaliate with proportional, temporary price cuts before gradually returning to the higher, cooperative price. That's a self-enforcing version of "tit-for-tat", exactly the mechanism the Evolution of Trust game was teaching you. And critically, this isn't a fragile fluke, it holds up even when firms have different costs, different demand, more than two players, and various kinds of uncertainty.